BOTA 271

Biology < Bios - life, logos - knowledge } study of living org.
- Physiology - function
- Histology - tissues
- Zoology
- Microbio - microbes
- Cell biology - cells
- Genetic - inheritance
- Botany

The main

✱ Why do we study plant physiology
Despite their immobility a great deal occurs in a plant life. eg water + discolved subs. move thro' special pathways within the plant from the soil.
A no. of chemical reactions occur.
✓ Photosynthesis
✓ Respiration

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Plant Physiology.

⇒ Study of plant function thro' the application of knowledge gained from modern physics & chemistry.
Plant physiologists:
- The fundamental unit of life in a plant is cells
- Organisms will comsist of various cells
- Cells are characteristed as special micromolecule.
These includes; ~~stach~~ starch - enegy & structural integrity.
- Nucleic acids - Contains genetic information.
- Proteins - Enzymes

Cells organise themselves into tissues → organs
i) Tissues includes; Epidermics, Vascular bundle.
ii) Organs includes; leaves, roots & stems

Chemical & Physical Properties of solutions & colloidal suspensions

Solutions;
H_2O → Universal solvent

Factors / properties of any water
✓ Size of the solute
✓ Effect of gravity

Photosynthesis
Mawalo

There's more a reason not to close he ones who make you

True soln
- Small
- No effect on the gravity to e mole
- Particles do not separate
- form homogenous soln with water

1) True solution

- Solute particles are small and comparable in size with water molecules (homogenous solution formed) eg salt, sugar sln
* No effect on the gravity to the molecules
The particles do not separate

Colloidal soln
- large in size
- form heterogenous sol
- Do no separate
- Bombarded by H_2O molecules

2) Colloidal solution

- Solute particles are large.
- Form heterogenous solution. eg clay particles
- They do no separate under the influence of gravity.
- They are bombarded by the water molecules.



3) Suspension or emulsion

- Solute part. are very large and cannot remain dispersed under the influence of gravity.
- Particles are solid - suspension
- Particles are in liquid state - Emulsion

* Most biological systems are existing in Colloidal.
They may be;
1) Hydrophobic - water hating - clay particles
2) Hydrophilic - water loving - starch

Most of biological systems (colloidal slns) occuring in organisms are hydrophobic solutions.

CHEMICALS OF LIFE

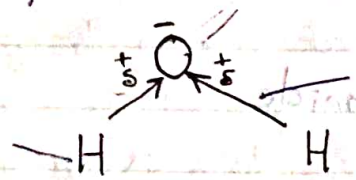
Topic 2

Chemicals of life

All living organisms have small building blocks that are either synthesised within the plants / acquired and are called Monomers.
- Monomers are linked together by strong covalent bonds.
- Monomers e.g Amino acids, sugars, fatty acids, acetic acid, glyceral, phosphoric acids, purins
- Polymer consisting of few monomers are called oligomer.
- Polymers could be straight, branched or even coiled into a rigid rod
- Sub-units (micro-molecules) of polymers are held together by electrostatic / Van der waals force - Responsible for most of biochemical properties.

Hydrogen Bond

✓ It is a non-covalent interaction.
✓ It is a special bond.



In the hydrogen atom the charge is shared in oxygen and it has partial positive & partial negative.

- A polarised bond due to charges being uneven.
- An individuals that have this polarised bond are called polar substan (soluble in water)
- Non polar e.g fats, hydrocarbons & other molecules of that nature. ie soluble in non-polar solvents.

Functional Groups

Most of biochemical rxns occurs in a portion while other part remains. It is a portion of a macromolecule that is involved in biochemical reactions.

- Examples are, Alcohol, Aldehydes, Ketones, Carboxyl group, Amines, Esters & Amide.
- X cal rxns are classified according to functional group that occurs.

Eg Esterification – Ester, Carboxylation – carboxyl

(functional g

i) CH_2-OH – Alcohol

ii) $-C(=O)-H$ – F.G – Aldehydes

iii) $CH_2-C(=O)-CH_2$ – Ketones

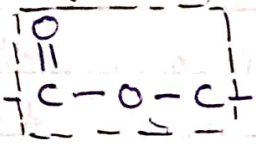
iv) $-C(=O)-OH$ – Carboxyl

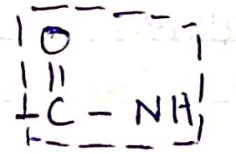
Carbon oxygen

Carbon – Nitrogen Group

i) $CH_2-CH_2-NH_2$ – Amino group.

Esters & Amide

i)  — Ester

ii)  — Amide

- Classification of carbohydrates

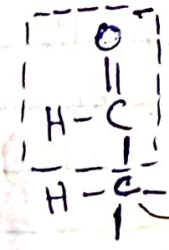
- Carbohydrates are groups of molecules composed of carbon, hydrogen & oxygen in an approximate ratio of 1:2:1
- It is classified into 4 ways;
  1) Based on No of sub-units
  ii) Monosaccharide, Disaccharides (2 mono), Oligosaccharide (12 mono)
  iii) Polysaccharide. (more than 12 monosaccharides)

Monosaccharide - one sub-unit

$(CH_2O)_n$

2. Based on the No of carbon units:
   a) Triose sugar - 3 carbon units
   b) Tetrose sugar - 4
   c) Pentose " - 5
   d) Hexose " - 6 carbon units.

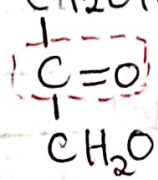
3) Based on the most oxidised functional group
   a) Ketoses - Ketone
   b) Aldoses - Aldehyde

ie.  - Aldehyde is most oxidised - Aldose

Assymetric carbon - bonded to 4 diff. atoms/grps

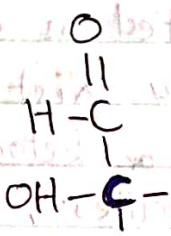
↓ Act as a fulcrum

D - configuration

ii)  - Ketone is most oxidised - Ketose

4. Based on the molecular symmetry

- Whenever 4 diff. groups are bonded to a central carbon atom the molecule is assymetric & the four groups can be arranged into two different configuration.



— Glycero-aldehyde

→ L-configuration

- The classification is either in D or L configuration.
- Two configurations have been designated as D or L forms.

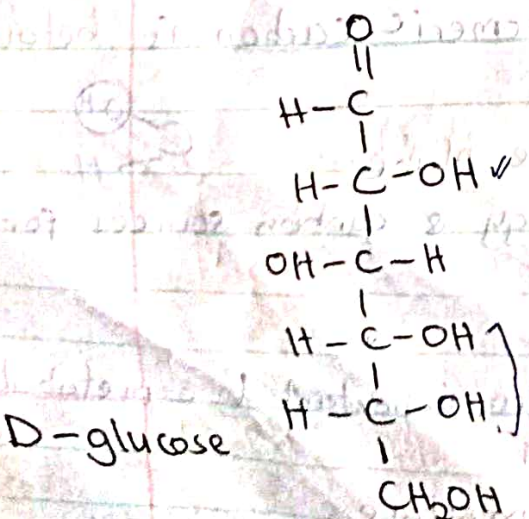
* The confi. of the assymetric carbon furthest from an aldehyde or ketone determine whether a monosaccharide belongs to D or L configuration.

* In the D-form the hydroxyl is on the right while in the L-form is on the left.

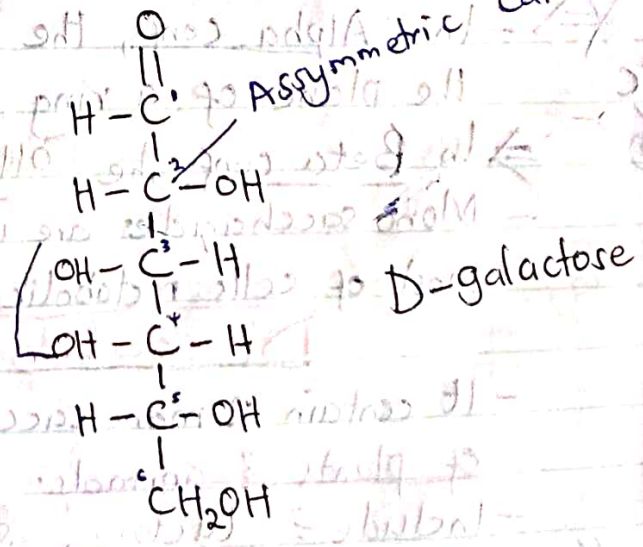
- Compounds with same chemical formula, but differ in position of hydroxyl grp on onel more assymmetric carbon are called stereoisomers.
- Stereoisomers that are mirror images of each other are known as enantiomers.
- Stereoisomers may also differ in position of hydroxyl group at only one assymetric carbon & are known as Epimers

eg D-galactose or D-glucose.

Examples of Epimers



D-glucose

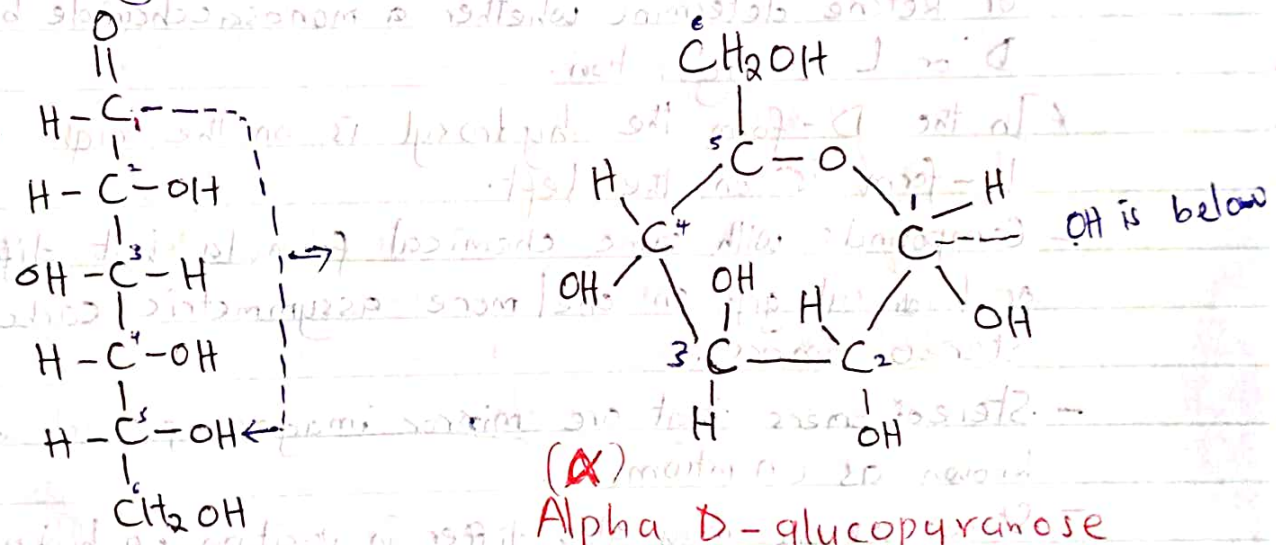


Assymmetric carbon

D-galactose

Structure of Carbohydrate

- Straight chain presentation of structure of a monosaccharide is called fisture projection formula.
- Ring chain/structure is called Harwolth projection formula
- Monosaccharides are often presented as straight chains (Fisture projection) however, they exist mainly in ring structures in which the aldehyde or ketone has reacted with hydroxyl group in same molecules.
- Ring sugars are common in sugar which contain 4 - 6 carbon atoms.
- Six ring structure is called pyranose.
- Five ring structure is called furanose.



(α) Alpha D-glucopyranose

⇒ Most oxidised carbon is called Anomeric / chiral carbon.
- The hydroxyl group in Anomeric carbon may be in Alpha / Beta configuration.

⇒ - In Alpha conf, the hydroxyl / anomeric carbon is below the plane of a ring structure



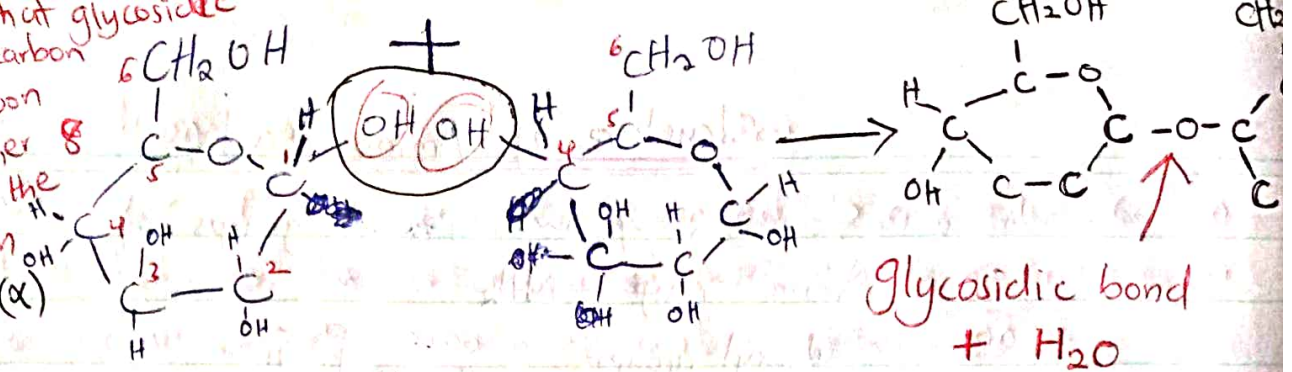
⇒ In Beta conf the OH is above the plane.
- Mono saccharides are important energy & carbon sources for most of cells metabolic activities.

Dissaccharide

- It contain 2 mono saccharides & also important to ē metabolism of plants & animals.
- Include : Lactose, Sucrose & maltase

⇒ The sugar units are joined by glucosidic bond & forms glycosid

α.1→4 indicates that glycosidic linkage connects carbon atom 1 with carbon atom 4 of the other & the config. about the anomeric carbon atom is alpha (α)



Maltase = glucose α.1→4 glucose

✓ Glycosidic bond is formed when OH on anomeric C. on monosacch. reacts with an OH of another monosaccharide. Glycoside can either be Alpha | Beta

✓ Alpha | Beta glycoside are ~~either~~ formed depending on position of atom attached to anometric carbon of the sugar.

Polysaccharide

-It include ÷ starch & cellulose which

Starch are cellulose are the most important polysaccharides

$$n(C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow (C_{12}H_{22}O_{11})_n + H_2O$$

✓✓ Starch is very importance as source of carbon and energy for germination ✓✓

Cellulose is the most abundant structural polymer

Important in plant cell walls and usually sub units are unbranched glucose molecules

Structure of Proteins

Are second Cellulose ÷

They consist of C, H, O & N Basically. Additionally some may have sulphur.

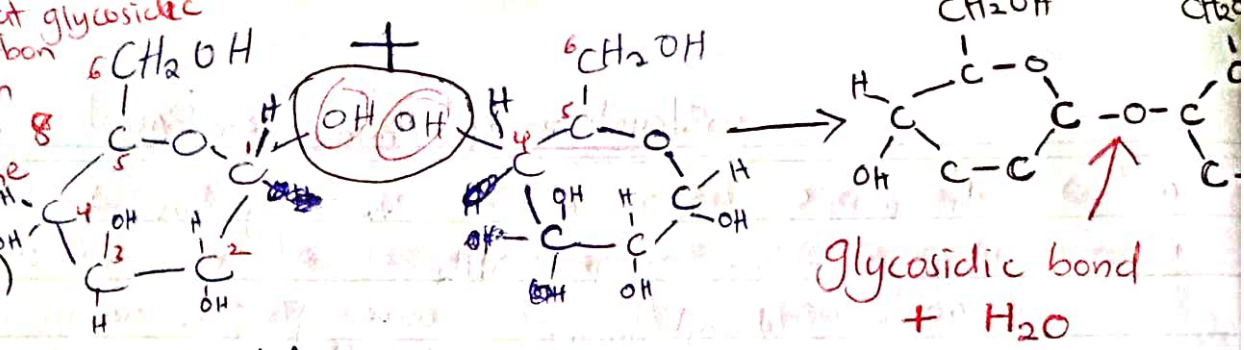
Function

→ Regulate chemical rxns - Enzymes

→ Make up of large portion of our cell membrane

→ Act as a storage of N in plants (Nitrogen)

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Structure of Amino Acids

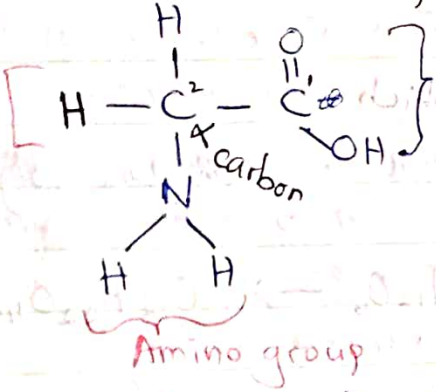
Each a.a. consists of two special groups of atoms

1- Amino group $-NH_2$

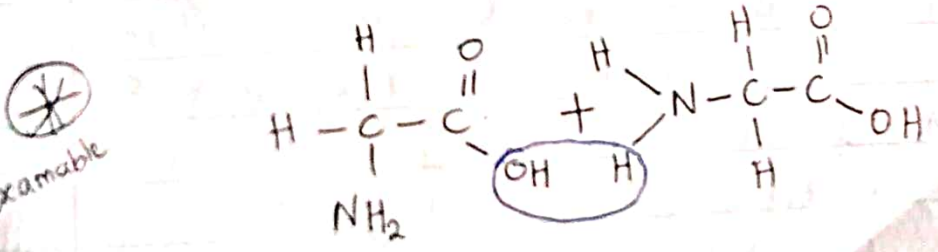
2- Carboxyl group $COOH$ 

3- Side chain - Varies from a single Hydrogen to a complex ring.

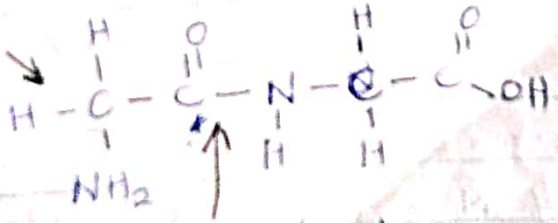
Basic structure of glycine



✓ Amino As are held together by peptide bonds. They are covalent bonds. They are formed btwn a reaction of OH on the carbon

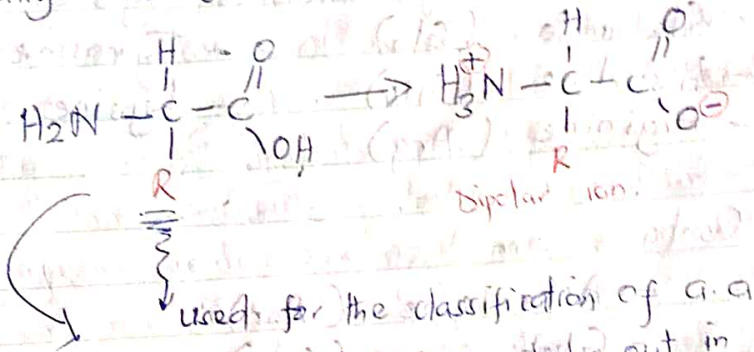


Dipeptide



Classification of Amino Acids

✓ Amino · A are also asymmetric with the D & L configurat
All ~~by~~ the 20 Amino · As except glycine are of ~~D~~ L conf
Beeos does not have an asymmetric carbon thence optically inactive



Used for the classification of a·a's
Usually larger and therefore sticks out in e structure and determine most physical & chemical properties

Groups of Amino acids

a) Hydrophobic amino acids
A group consisting of 2 sub-group
Has a side chain that can form hydrophobic
i) Aliphatic — Alanine (Ala) — Leucine (L)
— Valine (Val — Isoleucine (I
ii) Aromatic group — Phenyl alanine (Phe)
— Tyrosine (Tyr)
— Tryptophan (Trp)

b) Hydroxyl groups
This are ~~chi~~ side chain form hydrogen bonds
- Serine (Ser)
- Threonine (Thr)

(c) Sulphur Containing
Side Chains have sulphur
- Cysterine (Cys)
- Methronine (Met)

d) Ionisable groups

They carry a charge depending on pH. Form electro-static interaction

- Aspartate (Asp)
- Glutamate (Glu)
- Histidine (His)
- Arginine (Arg)
- Cysteine (Cys)
- Tyrosine (Tyr)
- Lysine (Lys)

e) Contains amides

- Asperagine (Asn)
- Glutamine (Gln)

f) Ring structure

- Proline (Pro)

A

Draw and ~~write~~ clearly identify the 19 amino acids with side chains other than hydrogen and classify them accordingly. (not contain glycine), (groups to be six), Histidine

Functions of Proteins

3 major functional groups -

i) Structural proteins
ii) Storage proteins
iii) Enzymes protein

i) Structural Proteins

Will occur in cell walls. Help to protect & prepare damage cells eg injuries. ✓ Cell expansion & growth importance.

ii) Storage Proteins

Accumulate in seeds and hydrolysed into their constituent amino-acids used for growth during germination.

(iii) Enzymes

Are large complex proteins that function as organic catalysts.

- Are pH specific & temperature sensitive.
- Break down bonds and allow for the formation of new ones.
- Speed up chemical reactions. (10^{20}) by lowering energy of activation of a substrates.

The minimum amount of energy needed to cause molecules to react - energy of activation.

*-stics ⇒ - They are extremely specific and they are able to differentiate btwn stereoisomers * filial isomers.
- They are required in very low concentration.
- They are reuseable.
- They are named according to the reaction they catalyse ending with a (Prefex (-ase)

Enzyme
- Maltose - Maltase
- Sucrose - Suerase
- Amylase - Amylase

Key
- uncatalysed
- catalysed

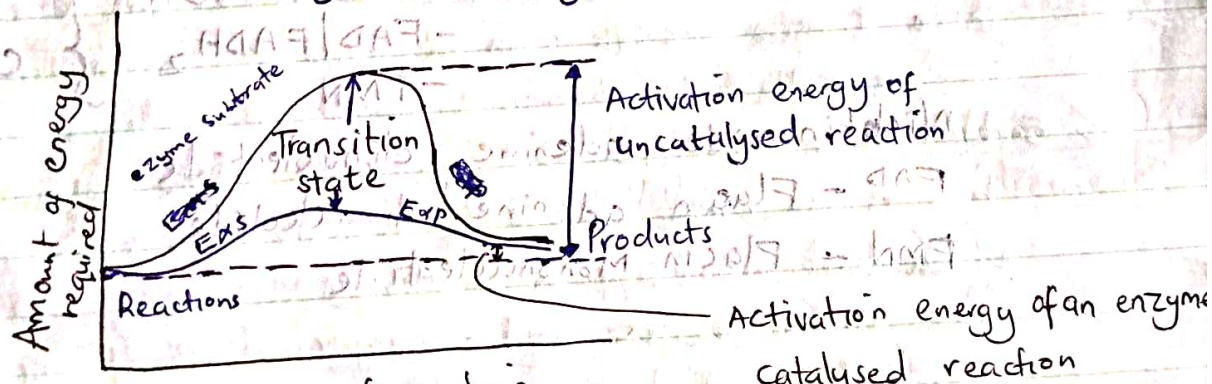


fig. The activation energy profile of an enzyme catalysed reaction Vs an uncatalysed reaction.

Enzymes ~~are~~ catalyse chemical reaction by temporarily binding substrate to its active site
A short ~~lase~~ link enzyme substrate complex is then formed as transition state is approached.
The transition state is an intermediate state btw the reactants & products where the necessary amount of energy and the correct arrangement ~~of~~ of atoms to yield the products are in place.

Behond the Transition st. another short leave enzyme produc complex exisit.

This complex thon breaks down as the products of reaction ar formed and the enzyme remains unchanged.

- Enzymes work with Co-factors.

Many enzymes require small organic molecules or co-factors that frequently are metal ions or derivatives of vitamins.

Co-factors (Prosthetic groups)

Co-factors are non-proteins molecules necessary for biological activities. Co-factor + enzyme → Responsible for specific nature of enzyme.

Co factors include;

- ✓ Metal ions - Zinc, Iron, Molybdenum etc
- ✓ Heme groups or Iron sulphur clusters
- ✓ Coenzymes - NAD^+ / NADH
  - $FAD / FADH_2$
  - FMN

} carriers

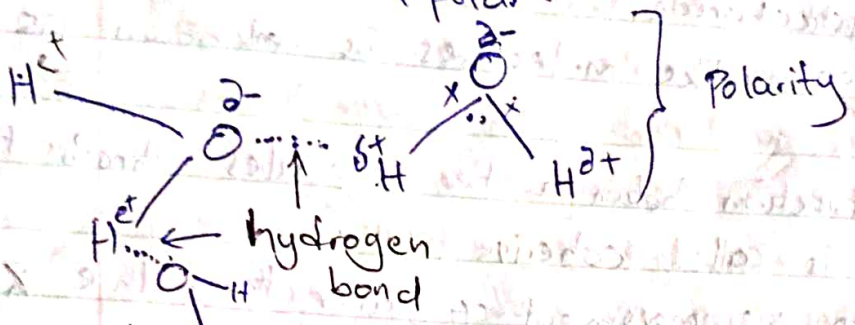
Nicotinamide adenine dinucleotide.
FAD - Flacin adenine dinucleotide
FMN - Flacin Mononucleotide

CHAPTER 3

PLANT WATER RELATIONS

PLANT WATER RELATIONS

- Water is universal solvent & important constituent to all living. This water is special due to its small size, polarity & presence of hydrogen bonds.
- Water is dipolar.



Water is liquid at room t. due to hydrogen bonds which prevents water molecule from escaping as water vapour.

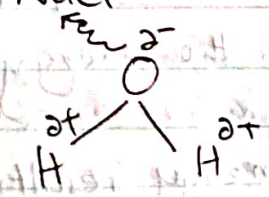
Chemical & Physical Properties of water

(1) Water is a solvent of both ionic & non-ionic substances (substances)

Why solvent - has high dielectric constant - able to diss both ionic & non-ionic substances.

Dielectric constant - Measure of the capacity to neutralise attraction btw charged particles

eg $NaCl_{solid} \xrightarrow{H_2O} Na^+ + Cl^-$ - more chemically reactive



(2) Water is incompressible liquid - important in plant structure cells turgid

Plant structure is dependent of sufficient structure exacted water on cell wall

Similary stomata respond according to the avail. of H_2O & opens when it moves in guard cells and but close when H_2O lost more to envir

(3) Adhesive & cohesive forces

Attraction between unlike substance

3. Adhesive - water & polar substances eg proteins
Cohesive force - water molecules attract each other
⇒ Becoz water molecule is polar it is attracted by other s
eg proteins & cell wall polysaccharide.
This attraction btwn unlike molecules is known as adhesive forces.
Similarity the attraction betwn water molecules through the formation of hydrogen bonds is called cohesive forces.
Both forces aid to the movement of water thro' the xylem vessels.

4. Has high tensile strength
⇒ This is the ability to resist a pulling force without breaking.
The existence of the cohesive forces makes water not to brea
(high tensile strength)
⇒ In a thin convined column of water eg in the xylem vessels
this tensile strength can be strong enough to enable water
to be pulled to the tops of the tall trees without breaking

5. Has high specific heat of vaporisation
⇒ This is the amnt of energy required to raise the temperature
of a unit mass of a substance by 1°C.
The high specific heat of vaporisation of liquid wate, is as
a result of a large proportion of the energy being used to
break the hydrogen bond.
Thus a large increase in heat energy result in a relatively
small rise in temperature.
This offers high temp. stability.

6. Has high latent Heat of vaporisation.
⇒ This is a measure of the energy required to vaporise a
liquid.
- To raise 1g of H_2O at 20° to 1g of H_2O vapor at 200°
requires an expendature of 2.45 KJ of energy there4
leaves dissipate excess each energy by transferring it to H_2O
The loss of water vapor result to the cobling effect

experienced by the leaves as they loss water by transpirat
7 Water is a reagent (Eg in photosynthesis it gives $\rightarrow H^+$)

TRANSPORT IN PLANTS

Transport Processes

1) Diffusion ⇒ Process thro' which molecules/substances move down a concentration gradient.

// Becoz molecules are in random motion (thermal agitation) they exchange kinetic energy. This energy propels substar move from one region 2 another.

The rate of transport of solute by diffusion is governed by Fick's 1st law of diffusion:

$J_s = - D_s \times \Delta C_s / \Delta x$ – Fick's 1st law of diffusion

where

J_s = Rate of solute transport = flux ≡ Amt of substa S, moving across a unit area per unit time.

- Unit = $(Mol\ m^{-2} s^{-1})$

Reduce D_s = Diffusion coefficient (Proportionality constant that relate the movement of a substance thro' various medium)

Increase ΔC_s = Concentration gradient (Difference in concentratio of substance S, in two regions

Δx = Distance b.twn the two points where the molecule are coming & going to.

Negative sign – This shows that the flow is down (decrease with time)

The difference in conc. gradient is the driving force for diffusio while the distance & diffusion gradient constitute for the resistance.

11) Bulk flow ⇒ (Mass flow) ⇒ Movement of groups of molecules in the response to a pressure gradient.

Cell contents experience pressure in the region of 0.4 – 0.5 M rigid cell wall and also by the diffusion process.

If a small incision is made on the cell wall its content will flow until there is pressure equilibrium.

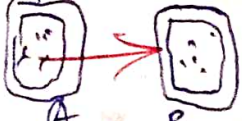
Bulk flow is the most predominant mean for long distance water movement via the xylem.
Unlike diffusion, it's independent of solute conc. gradient
(iii) Osmosis ⇒ The movement of a solvent thro' a selectively permeable membrane.

Osmosis can be considered as a special case of diffusion where water molecules move.
The process is a spontaneous & the source of energy is diffusion and bulk flow.
⇒ Movement of water from a region of high (wet soil) to low (dry soil) water potential with the release of free energy.
Water Potential is ~~denoted~~ denoted by the Greek letter Psi — ψ and the unit are Pa or MegaPa
The more negative Psi becomes, the more dry the soil is.
The lower the water potential the more the negative sign becomes & vice versa.
ψ (water potential) is derived from two constitutes

$$\psi = \text{Pressure potential} + \text{Solute potential}$$

$$\psi = P \pm \pi$$

ψ is generated when water molecules enters plant cells thro' osmosis making the cells turgid.
Osmotic potential / solute potential is caused by the presence of solute. Water potential (ψ) can be (+, 0, negative)
Solute potential (π) is negative or zero for pure water while the pressure potential (P) is usually positive.
Depending on the magnitude of its components water potential can be +, zero, -ve.
The values of P are always lower than those of π.
Becoz the values of P are normally lower than those of π, the water potential (ψ) of a solution is always negative.
In the soil, plants, atmosphere continuum water potential is highest at the soil and lowest at the atmosphere ~~and~~ with intermediate values in various parts of the plant.



P = 0.5 MPa
Π = -1.0 MPa

P = 0.3 MPa
Π = -1.5 MPa

} cellular poten

Determine the direction solution flow

Cell A $\Psi = P + \Pi$

$0.5 - 1.0 = -0.5$ MPa * water move from

B: Ψ $0.3 - 1.5 = -1.2$ MPa

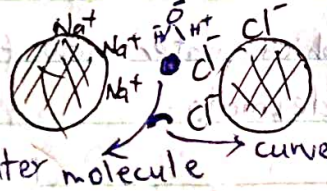
Uptake of water from the Soil

Soil water ~~Rate~~ flow is controlled by soil water potential (Ψ)

- Soil water potential is composed of osmotic pressure (Π) and hydrostatic pressure (P) (water content of so

Π is derived from dissolved solutes & is very low approx 0.01 MPa.

- P is dependent on water content on the soil.
- Soil osmotic pressure is generated by the dissolved salts and is generally low in the region of 0.01 MPa
- Hydrostatic pressure depends of the water content of the soil an less than or equal to Zero.
- Hydrostatic pre. develops becoz of the adhesive forces bidding water molecules to soil particles. This results in the creation a large S.A btwn the water and soil air



- Subsequently, the air-water surface stretches and develops a curved meniscus with a tension or negative pressure.

$$\text{Hydrostatic Pressure} = \frac{2T}{R}$$

T — Surface tension of water (7.2

R - Radius of the curvature.

- The hydrostatic pressure is generally greater than osmotic pressure.
- As the root hair absorbs water from its surrounding, it reduces the value of P in the water near the root surface.
- This establishes a pressure gradient becoz the neighbouring region has higher P values (high magnitude)
- Water moves this gradient to the root surface by bulk flow.

Hydrostatic Pressure is generally greater than atm. P, there, soil water is under great tension.

$$\psi_{soil} = \pi + \left(-\frac{2T}{r}\right)$$

$$\psi_{soil} = \pi - \frac{2T}{r}$$

As the root hairs absorbs water surroundings

Water moves down ie gradient to the root surface by

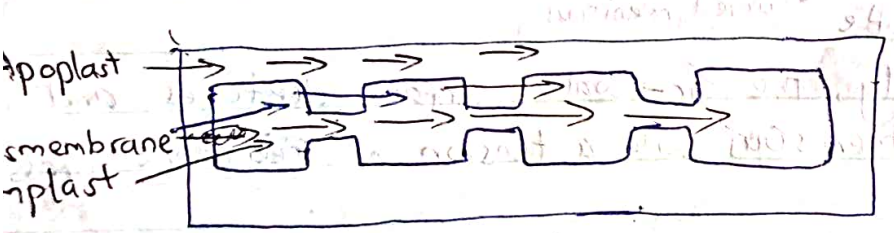
Pathways for water flow towards the xylem tissues

There are 3 ways upon the entery by root hairs, water flows the root epidermis to the endodermis thro' a combination of 3 ways.

a) Apoplast - water moves thro' cell walls without crossing any pathw
b) Transmembrane } vacular
c) Symplast } Cellular pathway

Transmembrane - water flows thro' the cells / membrane

Symplast - is a pathway thro' which water flow from one cell to another thro' plasmodesmata.



Water transport thro' xylem vessel

The xylem system account for 99.5% of water transport pathways thro' plants.
Xylem vessel generally consist of dead cells / tuber - they have no organelles / membrane
Xylem vessels have two types of water transporting cells

(i) Tracheids

They are elongated, spindle shape cells.
Their walls are highly lignified walls (Prevent water perforation)
They have numerous pits which are for devl of primary cell no secondary cell.
They are generally porous.

Tracheids

They are generally elogated spindle
- Highly lignified walls
- (11) Vessels
- They have all x-stics same as tracheids.
- At the end of vessels they have perforation.



They lack of cell membranes in the xylem tissues and pits perforations in the walls of this element allow water to move low resistance in respond to gradient.

The Cohension-Adhesion theory

The theory postulates that water at the top of trees develops large -ve hydrostatic pressure that pulls the water column in t xylem.



The -ve hydrostatic P develops at the surface of the cells ē cell walls in a leaf as water evaporates initially from the inter lar air spaces. Close to the mesophyles

- As more water diffuses out of the stomata what remains is draw close to the cell wall and the air-water interphase develops.

This reduces the water potentials in the cells close to the xylem vessels which draws water from the xylem sap with water pot

As the water is drawn from the leaf xylem vessel, tension is set u in the water column. This tension is transmitted down the stem xyle the roots by cohesion of water molecules.

Adhesive forces also tends to stick water molecules to the walls o the vessels.

Becoz water has a higher tensile strength, a large tension is required to break the column of water.

Combine the both forces (adhesive & cohesive), build up a force capable of pulling the entire water column upward by bulk flow

Limitations of the cohesive-tension theory

Translocation of organic solutes in the phloem

It is located on the outside of both primary & secondary vascular tissues.

Two reasons why sucrose is transported by phloem;

i) Sucrose is inactive

ii) Sucrose is highly soluble.

Unlike the movement of water thro' xylem, the translocation of sugars is bidirectional ie upward & downward.

Cells used for transporting phloem are sieve elements responsible for transport of sugars.

Sieve elements are living but they lack basic structures & organelles

Have companion cells responsible for most of metabolic activities

eg ATP synthesis, protein synthesis.

Mechanism of translocation in the phloem

Source (leaves) → Sink (storage or site of utilization.

Ringing experiment

Isotope labelling techniques

Gut content of feeding aphids

Factors that determine source-sink pattern of flow

1. Proximity of source to sink - Shorter one is easily mature to young developing leaves as well as the pit.
   - Lower leaves translocate their sucrose to the roots but not developing leaves.
2. Importance of the sink - Roots & shoot are given prefere for translocation of sugar early in life but thro' its dvlpt becomes a dorminant seed later.
3. Direct vascular connectivity btwn source & sink
   - If linked together, there is likelyhood of flow of photosynthes
   - Translocation of photosynthesis from the site of synthesis to the sink is a 3 step.

Steps

1. They have triose phosphate → its transported from the chloropl to the cytosol - conversion of sucrose.
2. The sucrose moves across 2-3 cells to the vicinity of the sink elements.
3. Active transport where sucrose is actively loaded into sieve eleme thro' a process called phloem loading.

- Exptal evidence shows that the flow rate of solute thro' the sieve elements in the phloem is in average of 1 meter per hour. (1m/h)

Pressure flow hypothesis (Munch's flow hypothesis)

⇒ This hypothesis is the most widely accepted as the depicting the most probable mechanism of phloem translocation.

- According to this hypothesis, the flow of soln in sieve element is given by an osmotically generated pressure gradient btw the source and the sink.
- Phloem sucrose is loaded and generate a high osmotic (solute) pressure in sieve element & there is steep drop in water potential.
- Then water enters the sieve element thro' process of osmosis to sought the problem of water potential & there is a rise in turgor pressure then desired pressure is generated.
- In sink, the sucrose is offloaded hence there is a lower osmotic

6 molecules of ribulose 1,5 biphosphate are generated molecules of GA3P.
3 molecules of GA3P are used to synthesis carbohydrates (glucose)
pressure & there is a rise in water potential which ensures that water exit in the phloem & there4 there is a decrease in turgor pressure. This is to create the osmotically generated water potential.
The contents of the sieve elements are there4 physically pushed along the translocation pathway by bulk flow.
backs of ptocure flow hypothesis: It has 3 stages;
Carboxylation of 6 molecules of CO_2 acceptor molecules, ribulose 1,5-biphosphate
n a molecule of 3 phosphoglyceter acid (3PGA)
eduction of the 12 molecules of 3PGA to 12 molecules of glyceraldehyde 3-
osphate (GA3P or triasephosphate)

PHOTOSYNTHESIS

- All life on earth depends on energy from sun. The most active photosynthetic tissue in higher plants (Mesophyll) within the mesophyll there is chloroplast (light harvesting pigment - Chlorophyll.
- Most harverting pigments are Chlorophyll A, Chlorophyll B & carotenoids. (PAR)
- Photosynthetically active radiation is the portion of light spectrum which is useful to chlorophyll.
- Through photosynthesis process plants oxidise water (with release of O_2) and reduce CO_2 into simple organic compounds (mainly carbohydrate), These are further transformed into proteins & lipids hence photosynthesis is an example of redox reaction.
Plants oxidise water using Photosynthetically Active Radiation (PAR) with release of CO_2
- Ribblose bisphosphate carboxylase - oxygenase (Rubisco) is the principle enzyme responsible for function of CO_2 in leaves & other green tissues. It is the most abundant enzyme on earth & may constitute 30% of protein leaves.
- The series of events that culminate in reduction of CO_2 involve the both light and dark reactions
- The reactions take place in the thylakoids (specialised membranes of chloroplast)
- Dark reactions occur in the stroma (the aqueous sorrounding the thylakoids)

GOD's mercy favour plan Grace will Bless

Dark & light reactions are involved in photosynthesis

Dark reaction occurs in the absence of light during the day

Light Reactions:

- takes places in the thylakoids (specialised membranes of a chloroplasts)
- Generate ATP & NADPH which is utilised during dark reaction

Stroma

Dark Reactions

- Occur in the stroma (aqueous surrounding of the thylakoids)
- CO_2 is fixed in the stoma using NADPH and ATP

Light absorbing units of the chloroplasts

Within the chloroplasts light energy is harvested by two diff. functional Units

- ✓ Photosystem I – 2nd
- ✓ Photosystem II – 1st

} Photosystems

Stroma

In the chloroplast light energy is harvested by two diffrn functional un known as photosystems.

These energy is used to power the transfer of electrons thro' a series electrons donors & acceptors.

The final electron acceptor in this series is NADP+ which is reduce to NADPH

Photosynthesis can be considered as a process as occuring in 3 levels/sc

- ✓ Cellular
- ✓ leaf level
- ✓ Canopy

At each level a potential max. rate exists (P_{max}) which is limited by environmental conditions of H_2O, temp & nitrogen

Grain crops synthesise their dry matter thro one of 2 major photosynthetic pathways: Ex C_3 – wheat, soybean, rice etc.

C_4 pathway – maize, sorghum, sugarcane, millet etc

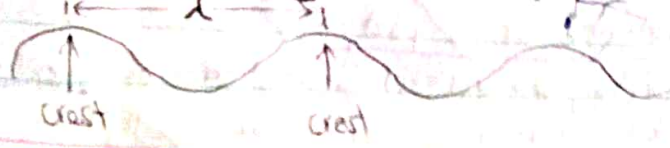
Light a driving energy source

Characteristics of light

Light has properties of particles & waves

Wave:

- ✓ Have a wavelength λ – distance btw successive wave crests
- ✓ Have a frequency ν – No of wave crests / unit time



$\frac{c}{\lambda} = \nu$ $\frac{E}{h} = \nu$ $\frac{c}{\lambda} = \frac{E}{h}$

Calculation of speed of light

($c = 3.0 \times 10^8\ m\,s^{-1}$) which is a product of its wavelength & frequency thus; $c = \lambda \cdot \nu$

As a particle: Light is called a **photon**.

- Quantum → energy contained in a photon of light

The energy is delivered in discrete amounts.

① The energy depends on frequency of light as given by Planks law thus; $E = h.\nu$ (h - Plank's constant ($6.63 \times 10^{-34}\ Js$))

② The energy of a photon is inversely related to the wavelength
✓ The shorter the wavelength, the more the energy.

Photon of blue light $\lambda = 400\ nm$ ($400 \times 10^{-9}\ m$) $= 4.97 \times 10^{-19}\ J$

Red light $\lambda = 680\ nm = 2.92 \times 10^{-19}\ J$

✓✓ Any energy btwn 400-700 nm is sufficient energy to start photosynthet process - PAR

Chlorophl $\xrightarrow{h\nu}$ Chl*

stable (ground state) — photon of light — excited - unstable

1. Re-emit ē photon - flourescence ✓✓
2. Convert ē energy of excitation into heat ✓✓
3. Photochemistry ✓✓

- When molecules absorb light they change their electronic state.
- Chlorophyl (Chl) absorbs a photon of light and moves to a higher or excited state (Chl*). In this excited state the distribution of electrons in the porphyrin ring changes.

✓ In the higher excited state chlorophyll is an extremely unstable and has 3 possible pathways to dispose off its available energy and return to the ground state. ✓✓ It can Re-emit the photon. ✓✓ (flourescence)

✓✓ It can also convert the excitation energy directly to heat ✓✓ without the release of a photon.

✓✓ Finally the energy of excited state can be used to cause chemical reactions to occur in a process of photochemistry. ✓✓

It occurs ~~occurs~~ more frequently bcoz ē rate of ē initial steps in ē photosynthetic process are among the fastest known chemical r

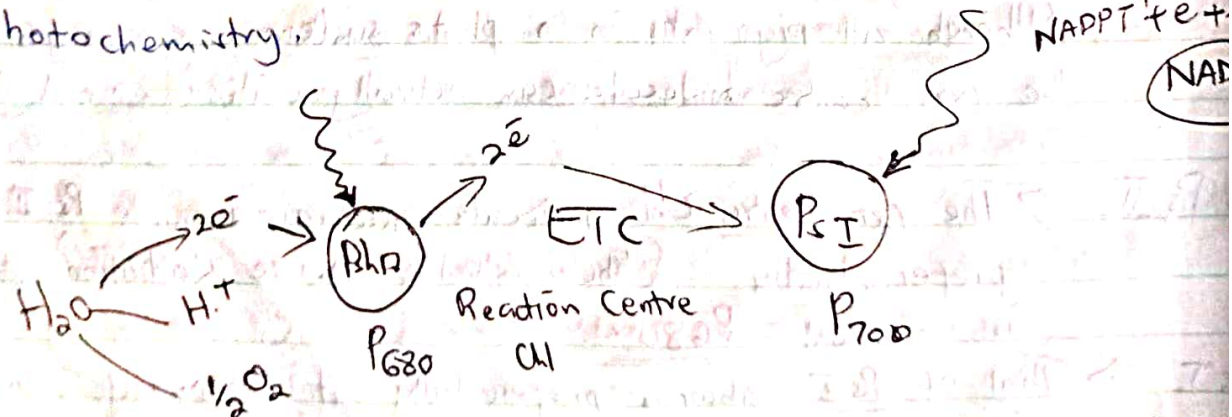
Light Absorption & Electron transport

22/01/

- Chlorophylls & bacteria chlorophylls are the light absorbing pig
- Both pigments have a complex porphyrin structure that contains some loosely held p electrons involved in electron transitions.
- Majority of the photosynthetic pigments serve as antennae col light and transferring the energy to a reaction centre where the che reactions which culliminate in energy storage occurs.
- Energy transfer is from a physical process while the reactions in the reaction centre are of a chemical nature.

antennae ⇒ Carotenoids are closely associated with both antennae and RC
RC pigment
photo-protection
centre pigments and are integral constituents of the photosyn tic membrane. They absorb light energy in the 400-500 band and rapidly transfer the energy to chlorophylls. Hence termed as accessory pigments.

- Carotenoids also play an important role in photo-protect because the photosynthetic membrane is prone to damage if larger amounts of energy are absorbed but cannot be ~~stord~~ stored in photochemistry.



- If the energy of the excited chlorophyll is not quickly utilised in photochemistry, a reaction occurs with molecular oxygen to form singlet oxygen.
- Singlet O_2 is extremely reactive and damages many cellular components.

1 468 000 000 400 000 000 Neymar Jr's month's salary

Carotenoids play their photo protective role by rapidly quenching the excited state of the chlorophyll.
- Further, the excited state of Carotenoid does not have sufficient energy to form singlet oxygen and decays to the ground state.

Light Dependente Reactions of Photosynthesis

ATP
NADPH |||| Light reactions

Photo system II (ChlA, ChB, Cart) $2\bar{e}$ → Elel Trac Chain → Photo s I (ChloA, Chl B, Carot) $\lambda = 700$ P_{700} (Chl$_{700}$)

$\bar{e}$

⇒ During the light dependent reactions of photosynthesis, protons are used to split water releasing electrons and protons & oxygen.
⇒ The electrons are used to generate via Phs II and Ps I the reducing agent NADPH and ATP for use in the light independent reactions (CO_2 fixation); Dark reaction
- Ps II & Ps I photosynthetic units are present in most chloroplasts
- Both are structurally distinct but linked by the $\bar{e}$ transport chain (ETC)
⇒ During photosynthesis, the events for Ps II occur earlier than Ps I
- Each photosystem consits of varying amounts of the major ligh harvesting pigments (ChL A & Chl B and carotenoids)
- Although all pigments in a photo system can absorb photons only the rxn centre molecule can actually utilise the light energy.

I ⇒ The rxn centre Chl molecule associated with Ps II absorbs PAR (Photosynthetic Active Radiation) at 680 nm and preferentially is denoted P680.
⇒ That of Ps I absorbs preferentially at 700nm and denoted P700
- The absorption of light by Chl 680 yields to give rise to an excited Chlorophyll molecule. donates an $\bar{e}$ to the electron transport chain (ETC) and generates an oxidising environment strong enough to split water.
- Similarly, ChlP700 absorbs light and gets excited (P_{700}^*)

Cont

and gains enough energy to allow the reduction of NADPH to NADPH.

These series of activities which represents the photosynthetic flow from water to $NADP^+$ is known as the Z-scheme photosynthesis.

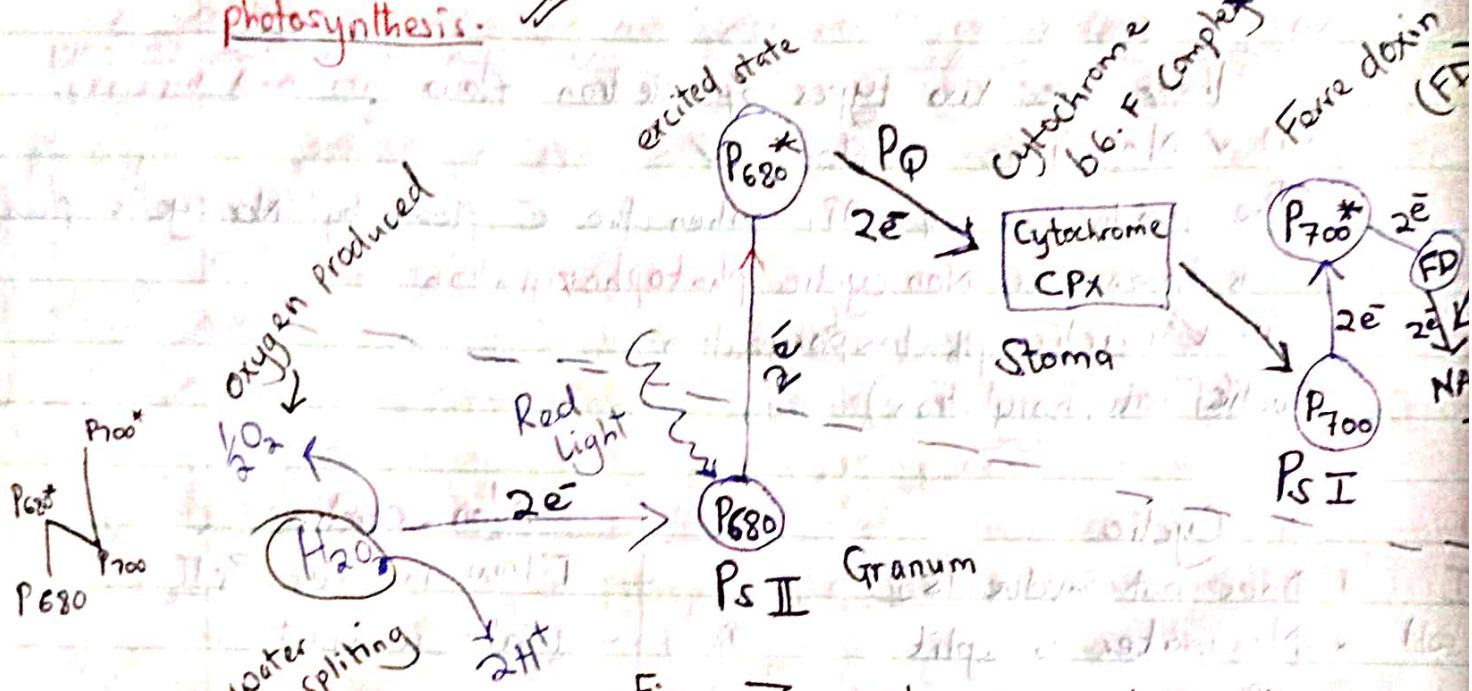


Fig. Z-scheme of photosynthesis

Electron flow and Photophosphorylation

$ADP + P \rightarrow ATP$ - high energy phosphate bond

$ADP \overset{\uparrow}{-} P$

$2H^+ \rightarrow$ Enzyme - ATP synthase

Chemiosmotic mechanism (Mitchell's chemiosmotic theory)

→ $\bar{e}$ potential diff $\bar{e}$

→ Protons → H^+

The Different between PS II and PS I

	Grana	Stoma	Net effect
1.	e^- produced	e^- released	e^- potential diff
2.	H^+ " in much quantities	H^+ produced in small quantities	H^+ differential

⇒ A part from PS II & PS I and cytochrome b of complex, ATP synthase is the other important cytochrome complex involved in the chemical rxns of photosynthesis.

$ADP + P_i \longrightarrow ATP$
photophosphorylation

Synthase

- This enzyme utilises the protons generated during the oxidation of water and those derived from cytochrome P6 complex to synthesises ATP
- Photophosphorylation works via a chemiosmotic mechanism which postulates that ion concentration difference & electrical Potential difference are source of free energy
- There are two types of electron flow utilised by cells.

(i) ✓ Noncyclic e^- flow ✓

⇒ The production of ATP when the e^- flow by Noncyclic flow is known as Non cyclic photophosrylation

(ii) ✓ Cyclic photosporylation

Cyclic (Photophorylation)

Cyclic	Non-cyclic
1 Does not involve Ps II	- Flow involves Ps II ✓
2 No water is split	- water is split. ✓
- Plants can utilise water when indefinite.	
3 No O_2 is produced.	- O_2 is produced. ✓
4 No NADPH is produced	- NADPH is produced. ✓

Photosynthetic Types (Dark reactions)

⇒ We have three

principle pathway
(i) Pre-dorminant - This is the C_3 photosynthetic pathway photosynthetic carbon Reduction (PCR) cycle

auxiliary dependent
(ii) C_4 photosynthetic pathway - (Carbon Assimulation (PCA) pathway)
(iii) C_2 photosynthetic pathway - (Photosynthetic Carbon Oxidation (PCO) pathway.

conjugated

(2) C_3 Pathway, PCR, Calvin cycle

- We combine CO_2 & water with a 5 carbon acceptor m

PCC

$CO_2 + H_2O \xrightarrow{\text{enzyme}}$ 5 carbon acceptor molecul

ATP

Regeneration

Reduced to $(CH_2O)_n$ ← 3 carbon intermediate

using NADPH + ATP → light reaction (stable product)

The PCR stages

We have 3 stages;

Carboxylation of 6 molecules of CO_2 to 4m 3PGA

(i) Carboxylation of 6 molecules of the CO_2 acceptor molecule ribulose 1, 5 - Bisphasphate (RuBP) to form 12 molecules of 3-Phosphoglyceric acid (3PGA)

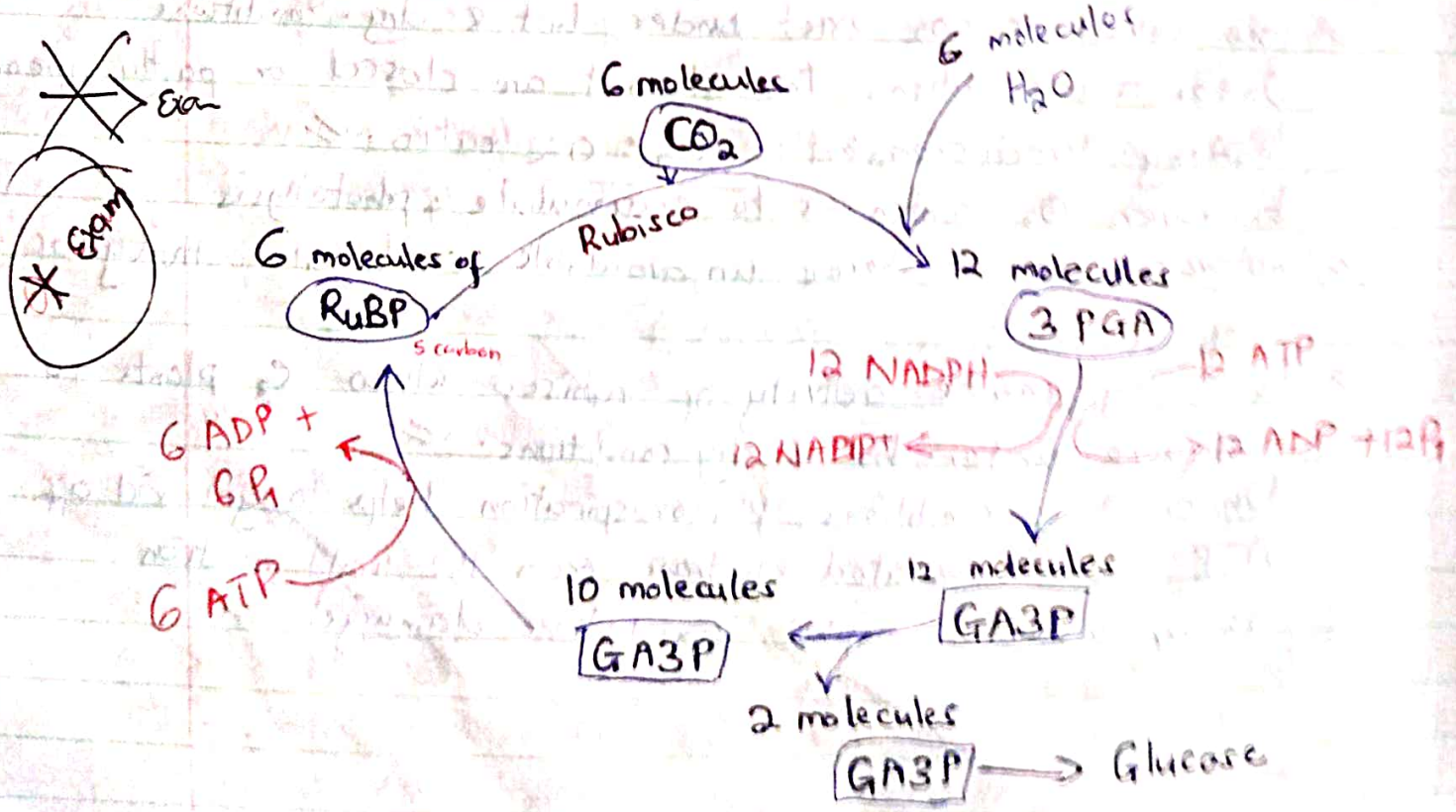
Reduction of 12 mole of 3PGA to 4m GA3P

(ii) Reduction step of 12 molecules of 3PGA into 12 molecules glyceralde hyde 3 phosphate (GA 3 P/triose Phosphate) (ATP + NADPH from light reactions)

(iii) 6 molecules of ribulose 1, 5 - bisphosphate are ~~regulated~~ regenerated from 10 molecules of GA3

Regeneration

Only 2 molecules of GA3P are used to synthesise $C_6H_{12}O_6$ (Glucose)



Fig; Principal steps of the PCR cycle;

Photorespiration

✓ CO_2 or O_2 depends on the concentration

✓ Rubisco is both - Carboxylase - can act with CO_2
- Oxygenase - O_2

✓ Rubisco has the potential to fix both CO_2 process of Carboxylate and O_2 and process is known oxygenation. ✓
✓ glycolate is formed

Photorespiration ⇒

✓ ⇒ Is a process thro' which rubisco fixes O_2 & CO_2 to release glycolate.

Situations where CO_2 becomes a problem

When it is hot & dry conditions CO_2 declines. ✓
✓ ⇒ The plants reduces their stomata sizes hence limiting water coming out like wise no CO_2 getting in.
⇒ When photosynthesis occurs O_2 increases but CO_2 declines

Photolysis (H_2O → $\frac{1}{2}O_2$, $H^+ \rightarrow ATP$, $e^- \rightarrow NADPH$)

⇒ At low CO_2 levels rubisco binds O_2 to Ribulose 1,5 Bisphosphate (RuBP)
✓ low levels of CO_2 exict under hot & dry conditions ✓
Under tric conditions the stomat are closed or partly opened leading to decreased CO_2 : O_2 ratio; ✓
However, O_2 continues to accumulate 4 photolysis
✓ Photorespiration is there4 un avoidable warteful activity of rubisco.

✓ The Oxygenate activity of rubisco allow C_3 plants to survive under hot & dry conditions. ✓
Under these conditions photorespiration helps to get rid off ATP and accumulated electrons from the light reaction. there4 preventing photo-oxidative damage. ✓

maize, millet, sugarcanes

(B) C_4 Pathway

- Found in crops in growing in tropics
- C_4 is an adaption to high temp and Irradiance (hot & dry environment)
- 0° ↑ temp ↓ solubility of CO_2

⇒ C_4 Reduces / eliminates photorespiration.

Adaptions of C_4 pathway

i) They have lower mesophyll resistance to CO_2 flow.
ii) There is CO_2 concentration within bundle sheath cells.
iii) CO_2 is delivered using Carriers to the site of Carboxylat
iv) It has CO_2 acceptor molecule with a higher affinity for
v) In C_4 pathway there is an additional chlorophyll containing cell
Bundle sheath / Kranz anatomy
vi) The C_4 pathway has a higher water use efficiency.

Steps / Stages of PCA (4)

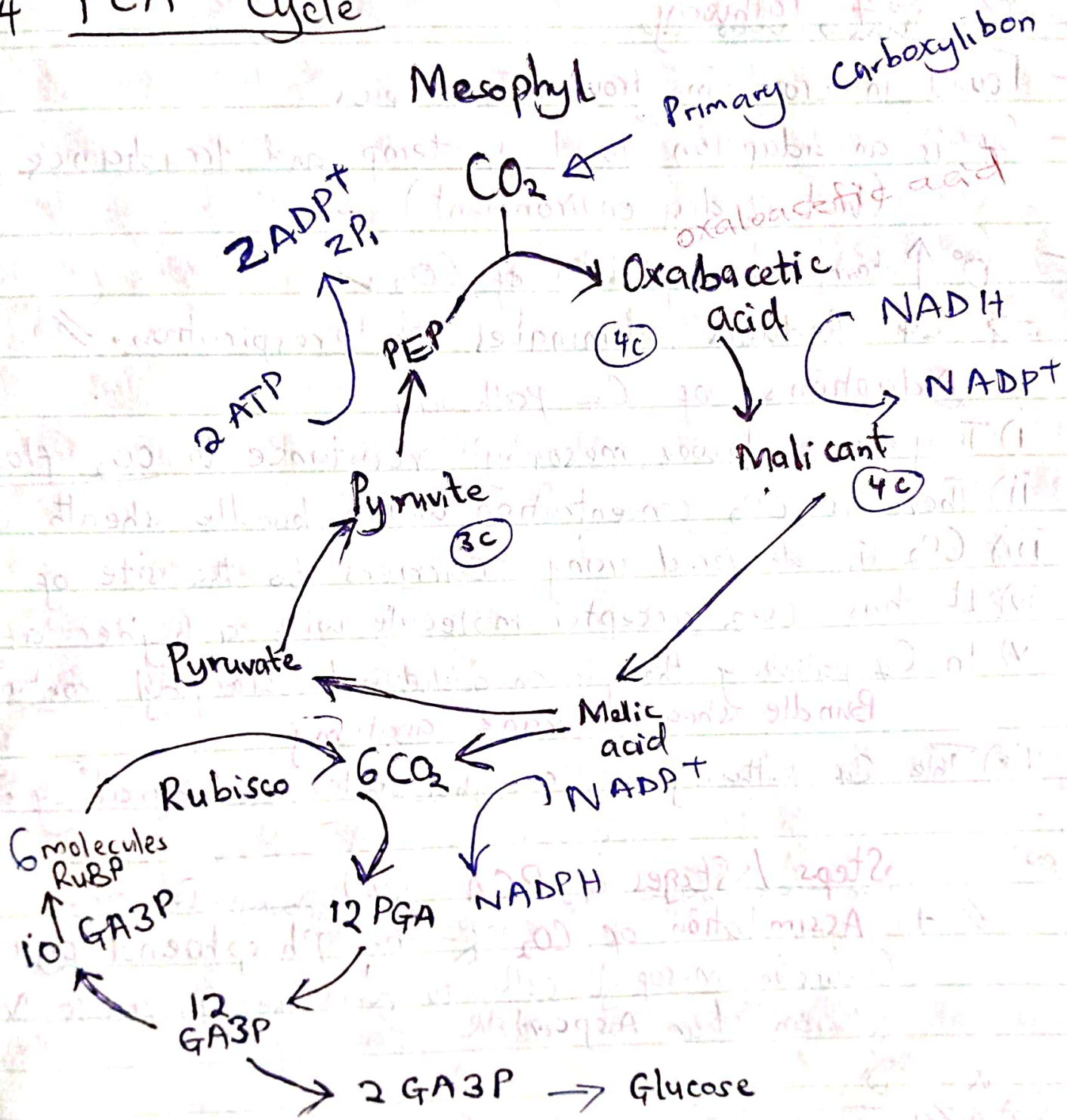
(a) Assimilation of CO_2 involving carboxylation of by Phosphoenol pyruvate (PE
Occurs in mesophyll cells to form the C_4 acids Malate and / or Aspartate.

(b) Transportation of the C_4 acids into the bundle sheath c

(c) Decarboxylation of the C_4 acids within the bundle she cells to generate CO_2 which enters the PCR

(d) Transport of the C_3 acids generated from decarboxylat which in most cases is Pyruvate or Alanine back to th mesophyll cells for regeneration of PEP.

C4 PCA Cycle



CO_2 conc. in the bundle sheath cells is achieved bcoz the 4 carbon atom is translocated from the many mesophyl cells to the

Under hot & dry conditions C4 are more effisient than C3 bcoz of many reasons:

* Read : Environmental factors that limit photosynthesis
✓ Temp ✓ water stresses ✓ Nitrogen deficiency

RESPIRATION

Introduction A part from photosy. respiration is further importa
energy and carbon tranforming system in plants.
✓ Oxidising energy rich molecules to give CO_2, H_2O, Energy
& heat. ← stored in4m of ATP
Cellular respiration (technically / technaly correct)
Chemically driven processes coupled with ATP synthesis
Start at Cytosol and ends at mitochondria.
Two type of resp:
Aerobic & Anaerobic → electron acceptor – other than O_2 (SO_4^{2-})
↳ flow of e⁻ and final product is O_2
In simplicity the illustration of the metabolism of glucose.
Sucrose or Starch → glucose + fructose → monomers of glucose

$$C_6H_{12}O_6 + 6O_2 \xrightarrow[\text{several steps}]{\text{enzyme}} 6CO_2 + 6H_2O + ATP + \text{heat}$$

Efficiency 35-50% energy
Other than ATP, carbon skeleton for biosynthesis of a.as, fatty acids & nucleic acids.

Major steps

a) Glycolysis
b) Tricarboxylic acid cycle (TCA), citric acid cycle / Krebs cycle
c) Electron transport chain

(a) Glycolysis pathway

This is the first major step of cellular respiration
It occurs in the Cytosol.
- The process requires No Oxygen
- 3 main steps and several smaller ones
- Each of these steps is catalysed by a specific enzyme
- Small amnt of energy is generated and transfered to ATP.

(1) Phosphorylation of the glucose molecule – 2 molecules of ATP are required.

- We get fructose which has 2 phosphates attached on either end of the glucose molecule = P — Fructose — P
- Fructose molecule is split into two, 3-carbon fragments
  Glycoraldelyde 3-phosphate (Triose Phosphate) (GA3P)

From Triase; hydrogen, water & energy removed → Pyrurate

⇒ A total of 4ATPs are synthesized during the process,

- Most of energy that was contained in glucose molecules is contained in Pyruvate
- The energy that is released is retained / held by NAD
- The hydrogen contains high energy and is contained in NADH $NAD^+ \rightarrow NADH$

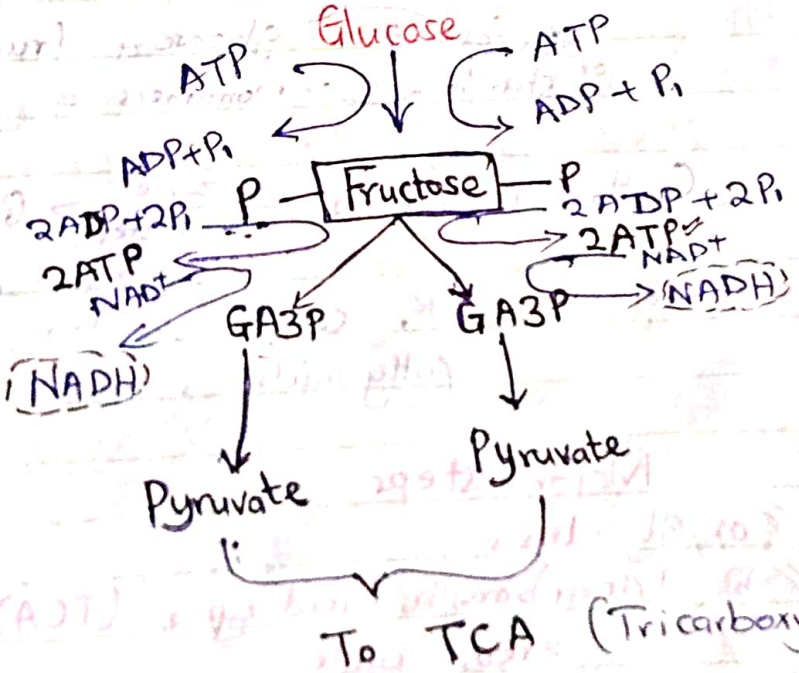


fig: An overview of glycolysis

The functions of glycolysis are produce ATP, Pyruvate NADA and carbon skeletons for biosynthesis

$$\text{Glucose} + 2\,NAD^+ + 2\,ADP + 2P_i = 2\,\text{pyruvate} + 2\,NADH + 2H^+ + 2H_2O + 2\,ATP$$

Aerobic Respiration

- Most common type of cellular resp. & occur in the Mitochondria
- It consists of Tricarboxylic acid (TCA) / citric acid / kreb's cycle / Electron transport chain.

Transformation of pyruvate before entering into the TCA

(i) Removal of CO_2

Then we get a remnant to a 2-carbon acetyl group.
The two carbon acetyl group combines with Coenzyme A to get
(2 carbon) Acetyl Co A which enters the TCA.

In the TCA acetyl CoA combines with Oxabucetic acid (4C) and Citric acid (6C) is given off.
We remove CO_2 (5 carbon remains)

NAD + H → NADH + H → 4C → CO_2
NADH → NAD+

Kreb's cycle

In Mitochondria

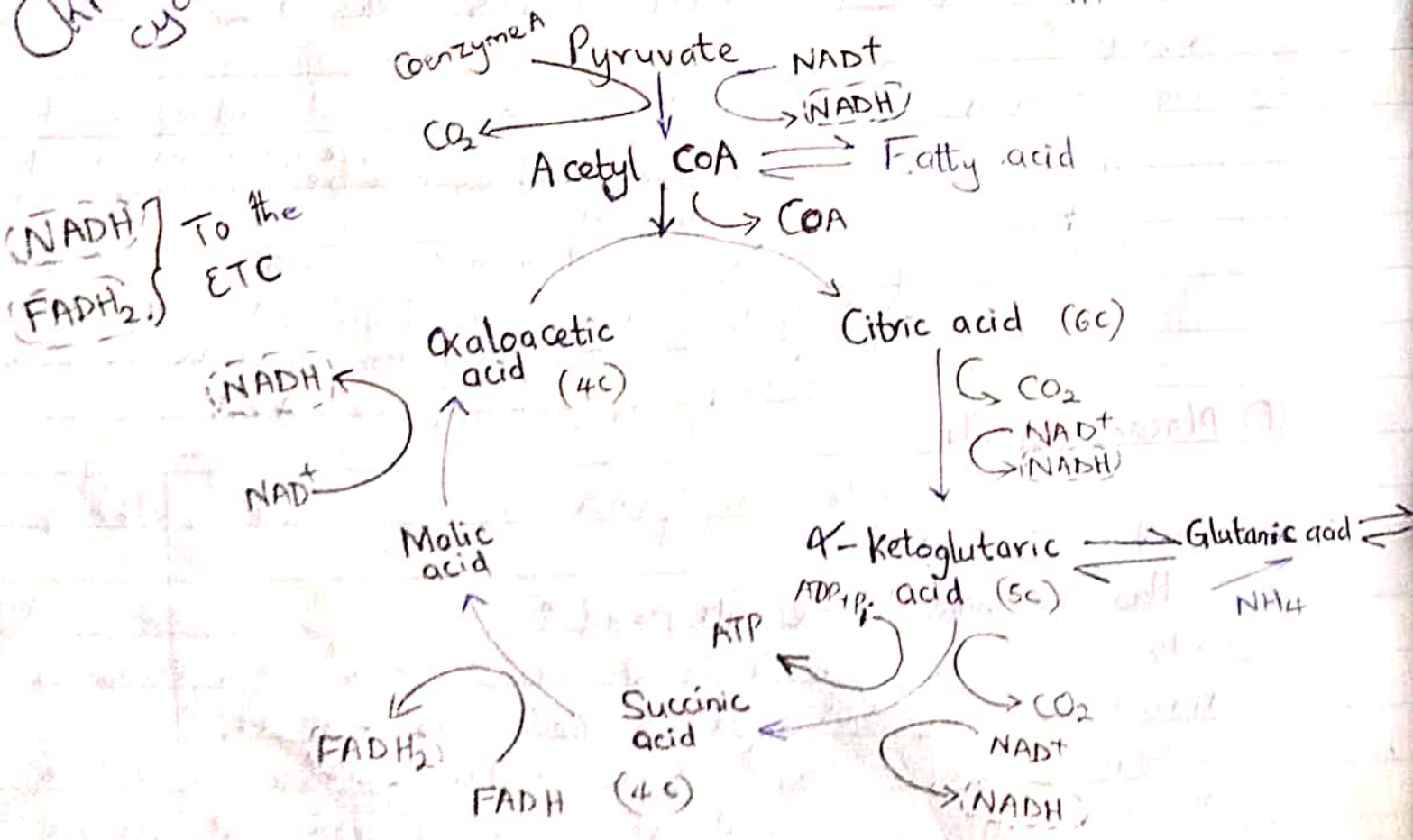


fig: An overview of the TCA

NOT produced ✓ fatty ✓ ATP ✓ Protein ✓ Pyruvate accur

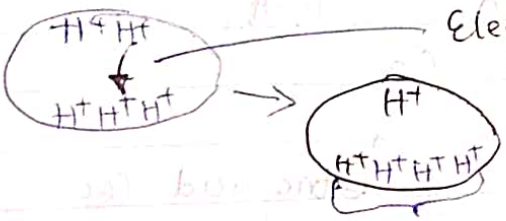
Oxaloacetic acid + Acetyl CoA + $3NAD^+ + FAD^+ + ADP + P_i$
= Oxaloacetic acid + CoA + $2CO_2$ + 3NADH + $3H^+$ + $FADH_2$ + ATP

The electron Transport Chain and oxidative phosphorylation

- Most of the energy in the reaction above is retained in the high electron carriers of NADH and $FADH_2$.
- These carriers cannot produce energy thus NADH & $FADH_2$ must go a series of oxidation reaction for the energy to resynthesis.
- The electron transport chain have several energy carriers which lead to the phosphorylation to ADP + ATP.

1 – As the ~~energy~~ e^- flow along to ETC there is generation of energy.

2 – The energy is used to shuttle protons from one end of mitochondria



Electrochemical gradient → serve as a source of ~~reserved~~ energy and through the process of Chemiosmosis electrochemical gradient results to phosphorylation of NAD which leads to synthesis of ATP.

(f) Placar deposits

* Minerals brought down by erosion and deposited as sediments

How are minerals deposits found ?

Finding a minerals deposits is the fist xted . The process is known as Mineral exploration which can be divided into 5 stages;

(1) Area selection; Potential means to deciding where to explore Desktop evalution of geological geochemical and geophysical data are conducted to found area that have potential to contain an ore body. Issues related to health, safety and env. & communities & technicalas or political ~~risks~~ risks are identified at this stage

Energetic Yield of Cellular Respiration

ATP Generated

	Direct	NADH	$FADH_2$	Sub total
Glycolysis	2	$2 \times 2 = 4$	—	6
TCA	$1 \times 2 = 2$	$2 \times 4 = 8$ $8 \times 3 = 24$	$2 \times 2 \neq 4$	30
			Total ATPs	36

Assumptions

1. 1 molecule of NADH from glycolysis thro' ETC ⟶ 2 ATPs
2. When 1 molecule of NADH from TCA thro' ETC ⟶ 3 ATPs
3. 1 molecule of $FADH_2$ from TCA " " ⟶ 2 ATPs

Minerals of either sedimentary

Classification

(a) Massive ore bodies – where there's considerable lateral & vertical development of the ore body and that the mineralisation is relatively uniform



(b) Bedded or tabular ore body



⇒ The lateral extend is wider than the vertical extend

(c) Narrow veins – This is the zone or belt of mineralisation typically long and narrow usually less than 10 feet width and often dipping steeply

(d) Wide veins – upto 100 feet

(e) Lenticular or pocket deposits – Minerals occur in lenses or isolated deposits



⇒ These are isolated small deposits in less forms